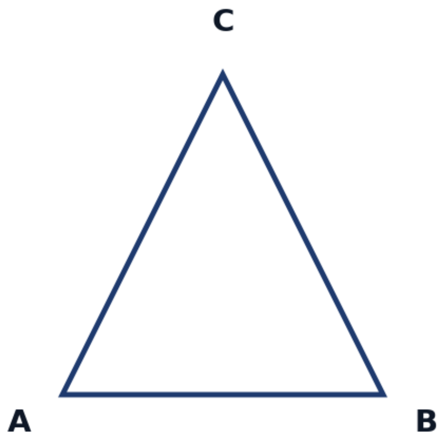
Advanced Coordinate Geometry

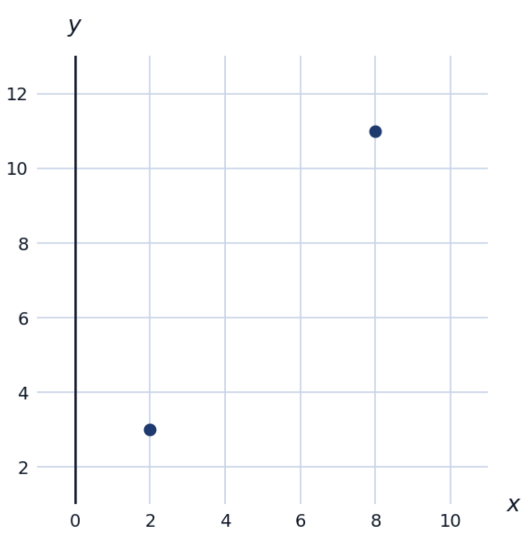
Distance, midpoint, and slope combined

1 Triangle ABC has vertices $A(-2, 1)$, $B(4, 1)$, and $C(1, 7)$. Find the length of the median from C to the midpoint of AB.



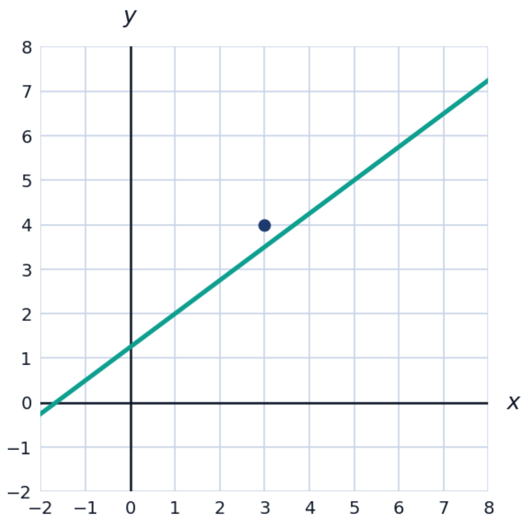
Equations of circles from given conditions

2 Find the equation of a circle with a diameter having endpoints $(2, 3)$ and $(8, 11)$.



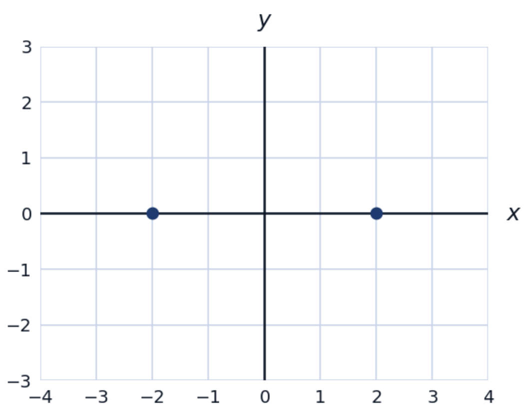
Distance from a point to a line

3 Find the distance from the point $(3, 4)$ to the line $3x - 4y + 5 = 0$, using $d = \dfrac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.



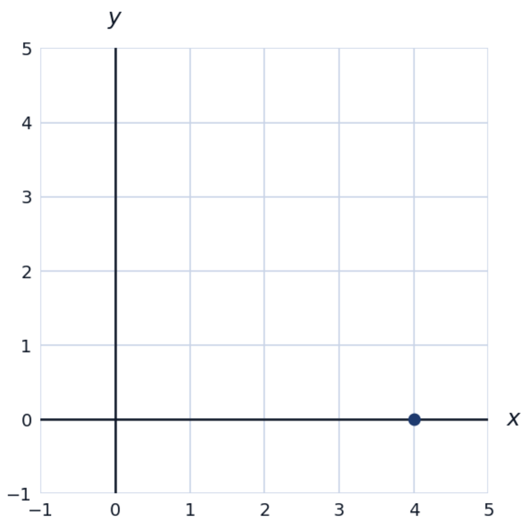
Locus problems

4 Find the equation of the locus of points equidistant from $(2, 0)$ and $(-2, 0)$.



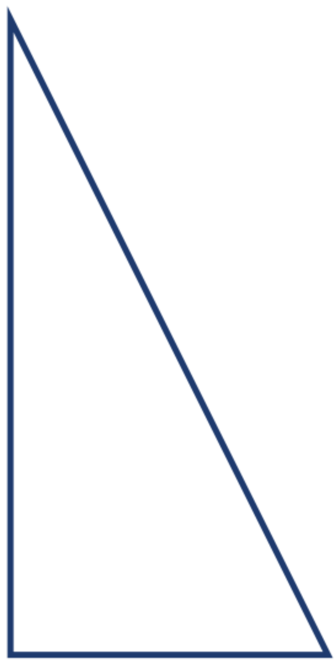
Rotations and general transformations

5 A point $(4, 0)$ is rotated 90° counterclockwise about the origin. Find the coordinates of the image point, using the rotation rule $(x, y) \rightarrow (-y, x)$.



Dilations and scale factor

6 A triangle with vertices $(1, 2)$, $(3, 2)$, $(1, 6)$ is dilated by a scale factor of 3 centered at the origin. Find the coordinates of the image vertices.



Finding the equation of a parabola from three points

7 A parabola $y = ax^2 + bx + c$ passes through $(0, 3)$, $(1, 6)$, and $(-1, 2)$. Find a, b, and c.

A multi-part triangle-in-the-plane problem

8 This question has two parts. Triangle PQR has $P(0, 0)$, $Q(6, 0)$, and $R(3, 6)$. (a) Find the area of the triangle using the coordinate formula $\text{Area} = \frac{1}{2}|x_P(y_Q - y_R) + x_Q(y_R - y_P) + x_R(y_P - y_Q)|$. (b) Verify the area using the base-height formula instead.

Finding the equation of a line given a point and a perpendicular condition

9 Find the equation of the perpendicular bisector of the segment connecting $(2, 3)$ and $(8, 7)$.

Translating and reflecting equations of curves

10 The graph of $y = x^2$ is reflected over the x-axis, then shifted 3 units up. Write the equation of the resulting graph.

Finding vertices of a polygon given transformations

11 A square has vertices $(0, 0)$, $(4, 0)$, $(4, 4)$, $(0, 4)$. Find the coordinates after translating the square 3 units right and 2 units down.

Finding the intersection of a line and a parabola

12 Find the points of intersection of $y = x^2 - 2x - 3$ and $y = x - 3$.

A multi-part quadrilateral classification problem

13 This question has three parts. Quadrilateral $WXYZ$ has vertices $W(0, 0)$, $X(6, 0)$, $Y(8, 4)$, $Z(2, 4)$. (a) Find the length of each side. (b) Find the slopes of WX and ZY, and of WZ and XY. (c) Based on parts (a) and (b), classify the quadrilateral as precisely as possible.

Finding the equation of a tangent line to a circle

14 Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$, using the fact that the tangent is perpendicular to the radius at that point.

Finding the vertex and axis of symmetry of a parabola from three points

15 A parabola passes through $(-1, 0)$, $(3, 0)$, and $(1, -8)$. Find the axis of symmetry and the vertex.